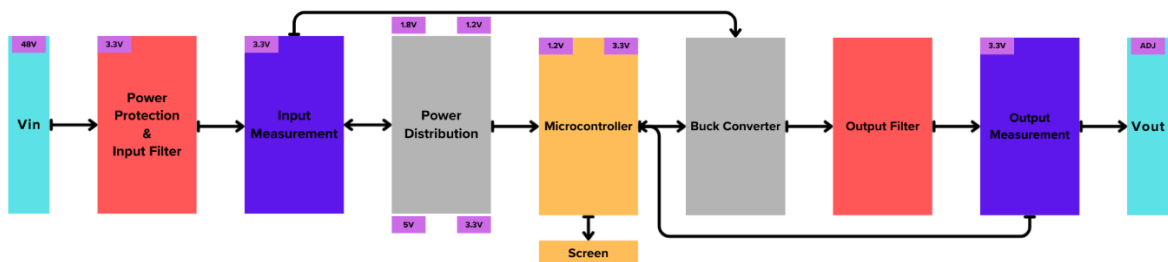


Designing with GaN – Miniaturization and Heating of High Frequency Converter

First Project Design Report

Project Purpose: Development of educational demo board with the possibility of observing by students the heating of the system elements with changes in parameters such as frequency, load, voltage and gain.

Stages 1: Circuit Parts & Block Diagram



- **Vin**
 - This part is just the power input of the circuit.
- **Power Protection & Input Filter**
 - This component protects the circuit from factors such as over/under voltage, short circuits, overcurrent, and inrush current. It also filters the input from factors such as noise.
- **Input Measurement**
 - This part measures the input voltage and current.
- **Power Distribution**
 - This part distributes the input voltage to our operating voltage levels.
- **Microcontroller**
 - This part controls and supervises all systems. It performs many tasks, such as measuring input/output voltage/current, switching the buck converter, controlling the display, and adjusting output variables.
- **Buck Converter:**
 - This part includes the buck converter topology
- **Output Filter**
 - This part creates the filter for the output after switching.
- **Output Measurement**
 - This part measures the output voltage and current.
- **Vout**
 - This part is just the power output of the circuit.

Stages 2: Determining Electrical Values

- $U_{IN} = 48V$
- $U_{OUT} = \text{Adjustable}$
- $F_{sw} = \text{adjustable}$
- $L =$
 - This value has a formula [1]: $\frac{U_{OUT} \times (U_{IN} - U_{OUT})}{\Delta I_L \times F_{Sw} \times U_{IN}}$
 - $\Delta I_L = 0.3 \times I_{OUT}$ and If our $U_{OUT} = 12V$ and $P_{OUT} = 100W$ then $I_{OUT} = 8.33A$ & $\Delta I_L = 2.5A$
 - Calculation factory of “0.3” is standard value.
 - Final calculation is $L = \frac{12V \times (48V - 12V)}{2.5A \times 500 KHz \times 48V} = 7.2\mu H$
- $C_{IN} =$
 - We need a bulk and filter capacitors for the input capacitors.
 - Bulk Capacitor Formula: $C = \frac{\Delta I \times \Delta t}{\Delta V}$, If our $\Delta I = 2A$, $\Delta t = 10\mu s$, $\Delta V = 1V$

$$C = \frac{2A \times 10\mu s}{1V} = 20\mu F$$
 - Input Filter Capacitors; It is aimed to create a wide-scale filter by adding **100 nF**, **1 μF** and **3.3 μF** capacitors to the input filter capacitors.
- $C_{OUT} =$
 - This value has a formula [2]: $\frac{\Delta I_L}{8 \times F_{Sw} \times \Delta V_{OUT}} = \frac{2.5A}{8 \times 500 KHz \times 0.1} = 6.25\mu F$

NOTE: When selecting capacitors, MLCC X7R low tolerance capacitor selections are priority.

Stages 3: Component Selections

- *Power Protection Parts*

We need a power protection part for our circuit against dangerous situations such as short circuit and overvoltage, so we will use the **TPS2492PWR** from Texas Instruments eFuse IC in this circuit.

- IC Calculations

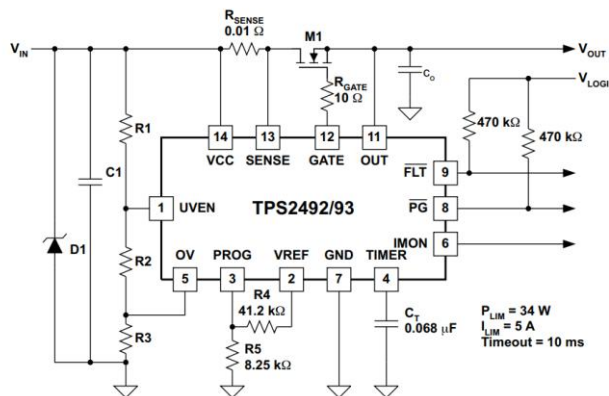


Figure [1] Typical Application Schematic

- $I_{LIM} = \frac{50 mV}{R_{SENSE}}$, If our current limit value is 10A, then R_{SENSE} value will be **5 mOhm**.
- Our threshold voltage value for the UVEN and OV pins is **1.35V**, so if our supply voltage range is **44V up to 52V**
 - $UVEN = 44V \times \frac{R2+R3}{R1+R2+R3}$, $OV = 52V \times \frac{R3}{R1+R2+R3}$ and $OV = UVEN = 1.35V$
 - Now our resistance values are **R1 = 374 kOhm**, **R2 = 1.82 kOhm**, **R3 = 10 kOhm**

- $V_{PROG} = \frac{P_{LIM}}{10 \times I_{LIM}}$, We need to apply a value bigger than 0.4 V to this pin, and when we calculate, our result will be 1.2 mV. Therefore, we can't use this pin. We need to connect PROG and VREF with a 47 kOhm resistor. (In this case, the PLIM value is our power lost in the MOSFET, not the total power loss of the circuit.)
- $C_T = \frac{36 \mu A}{4.1V} \times t_{ON} \times (1 + C_{O-TOL} + C_{T-TOL})$ & $t_{ON} = \frac{C_O \times V_{VCCMAX}}{I_{LIM}}$
 - C_{T-TOL} = Timer Capacitor Tolerance
 - C_{O-TOL} = Output Capacitor Tolerance
 - $t_{ON} = \frac{200 \mu F \times 48V}{10A} = 0.96 ms$
 - $C_T = \frac{36 \mu A}{4.1V} \times 0.96 ms \times (1 + 0.2 + 0.1) = 11nF$

Texas Instruments - F28P650DK6PZPR

- If we choose this MCU, It supports the high frequency PWM resolution. We can change the frequency and duty cycle with small movements.
- Also this MCU has a two core. We can easily control the screen with the other core.
- This MCU has a lots off PWM and ADC pins.
- Price is 14.71€ per unite / 10.10€ per unite (100 Quantity)
- GaNFET:
 - The EPC2306, EPC2307, EPC2308, and EPC2367 were found to meet the desired range and specifications. Based on differences such as R_{DS(on)} and size, it was decided to use **EPC2367**.
 - $V_{DS} = 100V$
 - $R_{DS(on)} = 1.2 mOhm$
 - $I_D = 101A$, Pulsed 420A
 - 3.3 * 3.3
- Temperature Sensor:
 - Texas Instruments - TMP119AIYBGR
 - It has high sensitivity and small size.
- Current Measurement:
 - Current Sense Amplifier: TI INA240
- Power Protection:
 - TPS2492PWR

References

- [1], [2] B. Hauke, "Basic Calculation of a Buck Converter's Power Stage," Dec. 2011. [Online]. Available: www.ti.com